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# Multi-Criteria Decision Analysis for the Evaluation of Nuclear Reactor Designs

Tesi di Laurea Magistrale in
Nuclear Engineering - Ingegneria Nucleare

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# Abstract

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# Abstract in lingua italiana

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**Parole chiave:** qui, vanno, le parole chiave, della tesi

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# Introduction

The global energy landscape is undergoing a signficant transformation, driven by the imperative to decarbonize electricity generation while ensuring energy security and system reliability. Within this context, the European energy transition is accellerating and nuclear power can play a critical role to both provide low carbon electricity and stabilize the grid amid the increase use of renewable power sources.

National requirments in terms of energy vary significantly, while some prioritize baseload capacity to offset import dependence, others require flexible solutions to be coupled with exisiting and future renewables. The emergence of Small Modular Reactors alongside established Generation III+ designs expands possibility but also introduces complexities to the decision making process. The committment to a specific nuclear reactor design involves complex tradeoffs across across technical, economic, regulatory, social, environmental, and geopolitical dimensions, which economic, social and political consequences will last for decades.

This thesis evaluates selected nuclear reactor designs for potential European deployment by employing Multi-Attribute Value Theory (MAVT), a structured decision-support methodology, and examines how different national contexts may influence the assessment process.

The research addresses four key questions:

- Which reactor designs are viable for European deployment based on geopolitical issues, technology maturity, and market availability?
- What are the key technical, economic and social attributes that differentiate the viable reactor designs?
- How do the energy requirments and socio-political contexts of different European countries influence these attributes?
- How does the application of a Multi-Attribute Value Theory (MAVT) framework reveal differences in reactor design rankings across countries?

# 1 | Literature Review

## 1.1. A brief History of Nuclear Power

### 1.1.1. Origins

The 1930s and 1940s witnessed rapid advancement in nuclear science. In 1932 James Chadwick discovered of the neutron [1] and a year later Leo Szilard hypothesized fission chain reactions. In 1938, Otto Hahn and Fritz Strassmann produced barium by bombarding uranium with neutrons, demonstrating nuclear fission [2]. Further advancements like the first controlled nuclear chain reaction in the Chicago Pile-1 achived by Enrico Fermi's team in 1942, were speed up by World War II as part of the Manhattan Project which lead to the atomic bombings of Hiroshima (6 August 1945) and Nagasaki (9 August 1945). The recognition among the scientific community of the peaceful potential of nuclear technology lead to the famous speech "Atoms for Peace" by USA President Eisenhower in 1953 to the United Nations [3] which established the foundation for international nuclear cooperation. The International Atomic Energy Agency (IAEA) and the European Atomic Energy Community (EURATOM) were established 4 years after, in 1957, and were tasked with promoting peaceful nuclear applications while serving as independent authorities for safety regulation and non-proliferation oversight.

### 1.1.2. Generation I

The 1950s marked the emergence of numerous reactor designs, driven by both civilian energy needs and military applications. These early models, now classified as Generation I reactors, represent the first prototypes of nuclear power at utility scale. The Experimental Breeder Reactor I (EBR-I) in Idaho (USA), which first generated electricity in 1951 (powering four light bulbs). The Obninsk Nuclear Power Plant (USSR), which in 1954 became the first to feed approximately 5 MWe into a local power grid. Calder Hall-1 (UK), the world's first commercial-scale nuclear power station, which began operation in 1956 with a 50 MWe capacity connected to the national grid

The United States pioneered light water reactor (LWR) technology through two distinct designs: the Boiling Water Reactor (BWR) and the Pressurized Water Reactor (PWR) which was originally developed for naval propulsion. The first commercial implementations of the BWR was the Dresden-I (1960) and for the PWR the Shippingport Atomic Power Station (1957) [4]. Concurrently, the USSR developed its own pressurized water technology, the VVER ("Water-Water Energetic Reactor"), featuring design characteristics that persist in modern variants.

This period also witnessed significant technological diversification as other nations developed their own reactor designs. The United Kingdom advanced gas-cooled Magnox reactors, optimized for dual electricity and plutonium production. The Soviet Union initiated development of the graphite-moderated, water-cooled RBMK design, while Canada's nuclear program focused on pressurized heavy water reactor (PHWR) technology, culminating in the first commercial CANDU reactor at Douglas Point in 1968 [4].

### 1.1.3. Generation II

Between the 60s and 70s Generation II reactors were deployed, taking advantage of the experience from the previous run, these reactors are not anymore considered experiments or prototypes but fully commercial power plants. In these decades numerous countries commissioned their first nuclear reactors: Belgium (1962), Italy (1963), Japan (1963), Sweden (1964), Switzerland (1969), Slovakia (1972), Argentina (1974), Bulgaria (1974) and Armenia (1976) [4]. Following the 1973 oil crisis, the deployment of nuclear power plants was accelerated in some countries as an energy security strategy. Most notably France implemented the Messmer Plan, named after the Prime Minister at the time, to greatly expand the nuclear plant fleet [5].

Nuclear expansion continued through the 1980s until 1986. As news from the Chernobyl disaster came to western countries, fear of the potential risks of nuclear power intensified. The public opposition grew significantly so that, even if safety and regulatory updates were implemented, several countries decided to slow down, reduce or halt their nuclear programs althogether, such was the case of Italy after the results of the 1987 referendum, which resulted in a complete phase-out of nuclear power [6, 4].

By the 1990s, multiple western countries had suspended nuclear deployment entirely, including Germany and Switzerland. France, previously the most active builder, connected 10 reactors between 1990-1999, while the United States and United Kingdom only deployed 4 and 1 reactors respectively during the same period. Italy completed its nuclear phase-out by 1990 with the permanent shutdown of all operating reactors [4].

Contrastingly, Eastern nations pursued an opposite trajectory. Following the post-Soviet economic transition, Russia resumed its nuclear program and has since continued expansion of its reactor fleet [4]. China initiated its nuclear program in the 1990s with Qinshan-1 (connected in 1991) and has since maintained aggressive expansion. Over the past three decades, China has brought 56 additional reactors online and currently has 29 under construction, positioning itself to surpass both France and the United States in total reactor capacity within the coming decade [4].

### 1.1.4. Generation III

As a result of the regulatory safety improvements, a stabilized choice of technology with LWRs settling as the to go choice all over the world and even larger capacities, Generation III reactor started to be developed in the 90s.

The improvement of these designs was fueled in the early 2000s by concerns over climate change, energy security, and rising fossil fuel prices, leading to Generation III+ reactors, such as the European Pressurized Reactor (EPR) or the US AP-1000 which improve safety and reliability in a consolidated plant layout. However after the 2011 Fukushima Daiichi accident Western nations experienced a significant setback: Germany accelerated its nuclear phase-out policy, while Italy reaffirmed its anti-nuclear position through a popular referendum [7]. In the following years projects faced substantial delays, cost overruns, and public opposition, with France's Flamanville-3 EPR serving as a particularly illustrative case experiencing budget increases to more then four times original estimates and construction timelines tripled from initial projections [8].

### 1.1.5. Generation IV

From the late 2000s to nowadays, new nuclear commissioning projects can be classified as "Mega Projects", such term is used to refer to very large committments where economies of scale tend to fail due to the complexity of the task at hand, resulting in massive delays and cost overruns, well known examples of Mega Projects are Suez and Panama Canals, the Sydeny Opera House or the development of the Concorde Supersonic Aeroplane [9, 10].

To counter this phenomena, the nuclear industry is developing Small Modular Reactors (SMRs) which are part of Generation IV reactors. SMRs are smaller, standardized reactors that can be built mostly off site to take advantage of economies of multiples. Some SMR designs are based on novelty technologies while others take advantage of well known water-based designs.

Other generation IV reactors are being design to tackle on other aspects of nuclear power such as waste minimization or higher efficiencies. Manly by emplying different means of cooling to exploit neutrons at higher energies (fast neutrons), or by using thorium instead of uranium as fuel or even using direct gas based cycles to increase thermal efficiency.

### 1.1.6. Current Landscape

This historical overview demonstrates the complexity behind the decision to adopt nuclear power which since it's beginning has been a matter of strategic geopolitical and historical consideration rather than purely technical or economic factors. So for starters we observed the current situation and consideration of nuclear energy in europe:

In 2020, the COVID-19 pandemic exposed vulnerabilities in global economic systems and Russia's invasion of Ukraine in 2022 further revealed Europe's energy dependencies [11, 12, 13]. At the same time, decarbonization goals to combat climate change have prompted Western nations to reconsider nuclear energy as a valid power source able to provide low-carbon, reliable power while enhancing energy independence and system resilience [14].

This strategic reassessment of nuclear is evident in recent policy initiatives. The European Union's classification of nuclear power as a sustainable investment under its taxonomy for climate mitigation [15], alongside programs like the U.S. Department of Energy's Advanced Reactor Demonstration Program [16], are a clear indicator of the renewed interest in nuclear technologies.

## 1.2. Countries

It is important to note that nuclear energy projects are charaterized by long development timelines from initial planning to commercial operation and can therefore face unique challenges related to political and social continuity. Frequent policy changes may introduce delays, require costly redesigns or halt projects althogether. For this reason the present study has considered four european country with different energy and political landscapes.

France demonstrates consistent public support for nuclear energy, with 75% favoring continued operation of existing plants and 65% supporting new construction according to a 2022 IFOP survey [17]. Given the long-standing consensus on nuclear power as a strategic energy pillar and the country's established industrial base, significant policy reversals appear unlikely in the near term.

Switzerland presents a more complex picture. Following the 2011 Fukushima accident, the government adopted a phase-out policy later confirmed by a 2017 referendum (42% approval) [18]. However, recent surveys indicate shifting public opinion, with 53% support [19, 20, 21]. The federal government has subsequently signaled potential reconsideration of nuclear's role in meeting climate targets [22], though the narrow margin of support and Switzerland's direct democracy system create policy uncertainty.

Poland shows strong public backing for nuclear development, with surveys reporting up to 92% support [23]. This aligns with concrete progress in the country's nuclear program, including preliminary site works and vendor agreements [24, 25], suggesting firm governmental commitment to nuclear deployment.

Italy's nuclear landscape remains constrained by the 1987 post-Chernobyl referendum and its 2011 reaffirmation following Fukushima, which mandated complete phase-out and new construction ban respectively. However, since the referendum in Italy are abrogative and do not need to another popular vote to change direction, a government decree is enough to reintroduce nuclear, which is what the current government has plans for [26]. Italy's history of political volatility and the long-term nature of nuclear projects create significant implementation challenges regardless of public opinion trends.

Author's Note: While public opinion surveys could provide, in principle, quantitative measures of nuclear support, closer examination often reveals methodological limitations that may skew results. In Italy, a 2024 IPSOS survey commissioned by Legambiente (an environmental NGO with a known anti-nuclear position) reported 81% opposition to nuclear power [27]. However, analysis of the survey report reveals problematic question design. The primary question was phrased as: *"The debate on nuclear energy has returned to the forefront. What is your opinion on the matter?"* with response options structured as:

- A1: "The current situation is complex; a return to nuclear should be evaluated"
- A2: "Nuclear has more risks than benefits and is not the right path"
- A3: "Nuclear energy entails excessive hidden costs and is not advantageous"
- A4: "Only safer technologies than current ones should be considered"

This framing demonstrates clear bias: two options (A2 and A3) explicitly oppose nuclear, A1 offers only conditional support, and A4 introduces safety concerns before suggesting potential support. Notably, the 81% opposition figure includes A4 responses, which represent conditional rather than absolute opposition. When considering only explicit opposition (A2+A3), the figure decreases to 43%.

A 2024 SWG study presents a more methodologically sound approach, though its conference presentation emphasized positive findings. The actual data reveals a more nuanced picture: over 80% of respondents consider themselves insufficiently informed about nuclear topics, and further investigation found that approximately 90% were unaware of SMRs or Generation IV reactors [28]. These findings suggest that public opinion on nuclear energy may be more influenced by lack of information than by firmly held positions.

## 1.3. MCDA

The present landscape presents policymakers with significant decision-making challenges. Renewed interest in nuclear power, geopolitical uncertainties and technological innovations, complicates the selection of an optimal reactor design. While nuclear commitments involve decade-long development timelines, 60-year operational lifespans, and substantial financial investments, modern energy policies must also consider evaluation of electricity demand, public dynamics, and supply chain vulnerabilities.

To address this complex decision-making environment, analytical frameworks have been developed, notably Multi-Criteria Decision Analysis (MCDA) which provides systematic support for evaluating alternatives when multiple, conflicting, criteria must be considered.

general description of the steps of the decision making process

## 1.4. Relevant Studies

A review of relevant literature reveals numerous applications of MCDA methods across diverse fields. Focusing specifically on studies combining MCDA with nuclear power decision-making, several distinct approaches emerge:

Site selection studies represent the most common application, typically employing GIS-based software to evaluate potential locations for new nuclear installations within limited geographical regions. These analyses frequently incorporate quantitative non-technical factors such as population proximity [29] [30]. Abdullah et al. (2025) extend this approach by including social dimensions like public acceptance and tourism impacts, which present challenges for quantification [31]. Notably, Devanand et al. (2019) develop an algebraic optimization model to identify optimal SMR locations in Singapore, though like most site-focused studies, they favor a design without sistematic comparison [32].

A second category compares different nuclear technologies. Kiser and Otero (2023) evaluates PWRs, SMRs, and Molten Salt Reactors [33], while Locatelli and Mancini (2011) compares large and small reactor designs [34]. Birikorang's study stands out for its regional focus on South Africa and comparison of three specific reactor models (NuScale SMR, EM2-HTGR, FTR-MSR), though its technological comparisons mix designs at different maturity levels [35]. Notably, all three studies utilize the Analytic Hierarchy Process (AHP) or its Fuzzy variant. The last one (Birikorang) draws on the IAEA's INPRO methodology for advanced nuclear technology comparison [36].

Broader energy comparison studies by Hernández-Torres (2025), Pasaoglu (2018), and Atilgan (2016) evaluate multiple generation options (typically eight alternatives) employing AHP-TOPSIS, AHP, and MAVT methodologies respectively. These studies use generic, averaged parameters for each technology class [37, 38, 39]. In contrast, Castelazo (2014) adopts a more precise approach using Multi-Attribute Value Theory (MAVT) to compare thirteen energy technologies, employing specific reference plants (e.g. EPR reactor for nuclear power) rather than generalized technology averages [40].

Three studies show particular relevance to the current research. Zolkaffly et al. (2014) evaluates nuclear plant models in Malaysia but suffers from outdated data and neglects non-technical aspects [41]. Topaloğlu (2025) applies the Analytic Network Process (ANP) to compare reactor types (PWR, BWR, PHWR) and locations, though social considerations remain limited to site-specific factors [42]. Goushis (2025) employs TOPSIS to assess SMR deployment in the EU for coal replacement, but focuses on a single reactor model rather than comparative technology evaluation [43].

The literature review reveals three gaps in nuclear energy decision-making research. First, most studies focus solely on site selection rather than comparing different nuclear technologies. Second, comparisons between nuclear designs lack considerration of a specific decision-making context. Third, studies that do consider specific decision-making context, examine various energy generation options as broad technology categories (e.g., nuclear, solar, wind) rather than conducting detailed analyses of specific reactor models or power plant designs.

## 1.5. Objectives

This study will consider real world data, literature estimates, industry claims and expert judgment to define indicators, thier value functions and weights. It evaluates six nuclear power plant designs to rank their sutability in four different european national contexts. In the first section the selectrion process of reactors to be evaluated is presented, followed by the choice of four countries. In the second section the definition of indicators, value functions and weights is discussed. The third section is dedicated to the final discussion

# 2 | Preparation

## 2.1. Criteria for Reactor Selection in Comparative Analysis

This study focuses on a curated set of reactor designs at an advanced stage of development. The selection process begins with the the IAEA "Nuclear Reactors in the World" [44], which catalogs operational, under-construction, and decommissioned reactors globally.

The initial selection process applied two primary exclusion criteria. First, designs were eliminated based on supplier origin, removing all reactors associated with countries unlikely to engage in nuclear technology transfer with Europe under the current geopolitical climate. This filter specifically excluded reactors developed in or supplied by Russia, China, and other nations maintaining restricted technology-sharing agreements with Western markets.

Second, reactors commissioned before 2000 were excluded from consideration, as their designs represent outdated technological standards. The complete filtering methodology is documented in the accompanying analysis notebook (PRIS Data Analysis.ipynb), which details how the initial dataset was reduced to nine distinct reactor models.

From this pre-selected group, three designs - the EPR-1750, APR-1400, and AP-1000 - were prioritized for comprehensive analysis. It should be noted that while the EPR (1650 MWe) and EPR-1750 share significant design similarities, they are treated as separate models in this study. Additionally, the Advanced Boiling Water Reactor (ABWR), though listed as operational in the IAEA ARIS database, was excluded following verification through the PRIS database, which indicates all ABWR units have suspended operation since 2011 [4]. The selection process and its outcomes are summarized in Table 2.1.

Subsequently, the IAEA Advanced Reactors Information System (ARIS) database [45], was consulted to identify advanced reactor models classified as under design, under regulatory review, under construction, or operational. Demonstrations units and prototypes were excluded from this set, yielding 10 candidate designs. To avoid redundancy, models already included in the prior selection (e.g., Generation III+ designs captured in both datasets) were removed. Each remaining design was then assessed for development status, technological readiness, and applicability to European energy systems as summarized in Table 2.2.

The overall selection produced a set of six reactors, three Generation III+ and three SMRs. Their main characteristics are summarized in Table 2.3. The most notable differences are the integrated designs of the BWRX-300 and Nuscale SMR which eliminate the external primary loop in favour of steam generator inside the reactor pressure vessel. This allows for compact construction and allows to exploit natural circulation for cooling. Then the Nuscale SMR is the only model designed to be deployed in a multi unit configuration, with up to 12 modules in the same containment building, for this study the 6 module plant (VOYGR-6) has been selected as it recently recived the Standard Design Approval by the U.S. Nuclear Regulatory Commission in May 2025. "A standard design approval indicates that a proposed reactor design meets applicable agency safety requirements. Companies that seek to use the US460 design would have to file applications seeking permission to build and operate a nuclear reactor using the approved design" [52].

Maybe here add a paragraph for each design to explain their main features.

| **Reactor Name** | **Country** | **Technology-sharing** | **Commercial Operation** | **Up to Date** | **Decision** |
|---|---|---|---|---|---|
| APR1400 | S.Korea | ✓ | ✓ | ✓ | **Selected** |
| CAREM Prototyp | Argentina | ✓ | ✗ | ✓ | Rejected |
| PRE KONVOI | Germany | ✓ | ✓ | ✗ | Rejected |
| EPR 1750 | France | ✓ | ✓ | ✓ | **Selected** |
| AP1000 | USA | ✓ | ✓ | ✓ | **Selected** |
| ABWR | Japan | ✓ | ✗ | ✓ | Rejected |
| APWR | Japan | ✓ | ✗ | ✓ | Rejected |
| EPR | France | ✓ | ✓ | ✗ | Rejected |
| OPR-1000 | S.Korea | ✓ | ✓ | ✗ | Rejected |

Table 2.1: Selection of reactor models for analysis

| Reactor Name | Country | Used | Reasoning |
|---|---|---|---|
| APR+ | S.Korea | ✗ | As reported in the manufacturer website [46] this design is mean to be an "indigenous reactor cleared of obstacles to export" |
| ATMEA | Japan, France | ✗ | Joint venture was abandoned in 2019 [47] |
| BWRX300 | USA, Canada | ✓ | Design from GE-Hitachi, a lot of support from US and Canada as well as interes in the EU [48] [49]. |
| i-SMR | S.Korea | ✗ | Never expressed interest in development outside south Asia and the middle east, no updates since early 2024. |
| iMSR400 | US | ✗ | No interest overseas, project seems to be at an early stage of development. |
| Natrium | US | ✗ | Demo plant [50]. |
| Nuward | France | ✗ | After the change in direction of 2024 the project has been set back [47]. |
| Nuscale | US | ✓ | Even if there was some financial controversy in 2024 [47]. the project seems to be still moving forward in terms of goverment founding [47] and regulatory review [47]. |
| PRISM | US | ✗ | GE-Hitachi considers it a "concept ... currently being put into practice in two reactors: Natrium and ARC-100" [47]. |
| Rolls Royce SMR | UK | ✓ | It's moving forward and from the technological point of view this is the simples as it it litteraly a scaled down PWR[47]. |
| SMART | S.Korea, Saudi Arabia | ✗ | Plan to export only to south-east asia and middle east [51]. |

Table 2.2: Selection of advanced reactor models for analysis

| Reactor Name | Type | Construction | $P_e$ (MWe) | Supplier |
|---|---|---|---|---|
| APR1400 | PWR | 1 Unit | 1340 | KEPCO (S.Korea) |
| EPR 1750 | PWR | 1 Unit | 1630 | Framatome (France) |
| AP1000 | PWR | 1 Unit | 1120 | Westinghouse (USA) |
| BWRX300 | i-BWR | 1 Unit | 300 | GE-Hitachi (USA-Canada) |
| Rolls Royce SMR | PWR | 4 to 12 Units | 470 | Rolls Royce (UK) |
| Nuscale | i-PWR | 1 Unit | 77 | Nuscale Power (USA) |

Table 2.3: Final selection of reactor models for analysis

## 2.2. Country Selection for Comparative Analysis

This study evaluates four European countries: France, Italy, Poland and Switzerland, to assess how national energy contexts influence the ranking of nuclear reactor designs.

In Italy, the energy market is divided into different zones, each with its own demand patterns. For this analysis, Northern Italy serves as the focal region being the country's most industrialized and energy-intensive area[47]. It is characterized by exceptionally high electricity demand, a strong reliance on imports, and above-average greenhouse gas emission intensity, making it a critical bottleneck in Italy's decarbonization efforts. Additionally, the country's historical political volatility introduces uncertainty into long-term energy planning.

France, with $\sim 70\%$ of its electricity generated by nuclear power[47], represents a mature nuclear market. Here, the marginal gains from additional reactors are likely minimal, offering insight into how reactor rankings shift in a saturated nuclear energy landscape.

Poland, heavily dependent on coal and among Europe's highest GHG emitters [47], presents an opposite case. As the country seeks to modernize its grid and reduce emissions through nuclear energy [47], it provides an opportunity to assess which reactor designs could best support its transition while meeting growing energy demand.

Finally, Switzerland provides a unique case due to its low electricity demand, hydropower-dominated grid, and prudent nuclear policy. Its constrained grid capacity and emphasis on energy independence make it ideal for assessing small modular reactors (SMRs) and their role in complementing renewables.

## 2.3. Attribute Definition

Throughout this section we will define the indicators.

### 2.3.1. Technical parameters

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### 2.3.2. MOX Fuel Compatibility

Mixed Oxide (MOX) fuel is produced through spent nuclear fuel reprocessing and offers a solution for partially closing the nuclear fuel cycle in thermal reactors. While this process generates some secondary waste, it provides several strategic advantages: reducing final nuclear waste volumes, minimizing mining waste, and decreasing dependence on uranium imports. The latter is particularly relevant for Europe, which lacks significant domestic uranium deposits. However, reprocessing also presents proliferation risks since the extracted plutonium could potentially be diverted for weapons production. Due to these competing factors, nuclear fuel reprocessing is a fundamental strategic decision; the United States has foregone this practice, while France and UK have implemented the process in their nuclear programs [53].

Given the European context of this study, MOX compatibility is considered a relevant reactor attribute. The evaluation methodology assigned each reactor design an initial score (0-10) based on its maximum MOX core fraction capability, followed by qualitative adjustments reflecting design-specific features. The complete scoring process and results are detailed in Table 2.4.

| **Reactor Model** | **MOX[%]** | **Considerations** | **Score** | Sources |
|---|---|---|---|---|
| APR1400 | 33 | - | 3/10 | [45] |
| EPR 1750 | 100 | 50% without modifications, to reach 100% "design adaptations which are not significantly costly" are required. | 9/10 | [45, 54] |
| AP1000 | 50 | 50% without modification, to reach 100% major modifications are required | 6/10 | [45, 55] |
| BWRX-300 | 0 | GNF-2 Not manufactured with MOX | 0/10 | [56] |
| Rolls Royce SMR | 0 | Rolls Royce states that they are not planning to use MOX | 0/10 | [57] |
| NuScale | 0 | Not licensing for MOX but a study reveals support is possible without design modifications | 1/10 | [58, 59] |

Table 2.4: MOX fuel compatibility scores and maximum MOX fraction

### 2.3.3. Technological Readiness

Expert consultation

### 2.3.4. Safety

To be discussed

### 2.3.5. Capital Costs

DRAFT

| Option | PROs | CONs |
|---|---|---|
| Use NOAK estimates | Use of estimates in all cases, comparable values | It's not realistic and not fair towards more mature designs |
| Use FOAK estimates/data | Better data for three large reactors | We are mixing up real data and estimates |
| Use NOAK and apply FOAK-to-NOAK conversion | Comparable values, with awareness of current know-how | rely on a model |
| Present NOAK and FOAK to experts along with technological readiness overview and let them evaluate | As before | rely on expert judgment |

Table 2.5: Options overview

The same applies to commissioning time.

It is important to note that these costs refer to an Nth of a kind installation and will not be representative of the FOAK costs. Moreover as detailed by Rothwell 2022 [60] the costs of recent nuclear installations in Europe and the US have been significantly higher than those announced at the start of the project. For this reason we will apply a FOAK-to-NOAK factor to better represent the real costs. For example the EPR-1600 announced costs for Flamanville-3 was 1886 $USD_{2020}/kW_e$ but have risen to 8600 $USD_{2020}/kW_e$ [61]. All Values are summarized in Table 2.6.

For reference, the capital costs of the same plants as the previous section have been reported by the US Energy Information Administration [62].

We want to clarify that the data has all been converted to 2020 USD since the main source of data [63] and [61] reports all costs in this currency and year.

| **Reactor Model** | **Capital Cost of NOAK [$/kWe]** | **Capital Cost of FOAK [$/kWe]** | Data Source |
|---|---|---|---|
| APR-1400 | 1828 | 2410 | [63, 61] |
| EPR-1750 | 2150 | 8600 | [63, 61] |
| AP-1000 | 2000 | 8600 | [63, 61] |
| BWRX-300 | 2318 | 3467 | [64] |
| Rolls Royce SMR | 6050 | 7408 | [65] |
| NuScale | 2850 | 3600 | [66] |
| **Other Power Plants** | | | |
| Onshore Wind | - | 1265 | [62] |
| Solar PV + Battery | - | 2175 | [62] |
| Hydroelectric | - | 6007 | [62] |

Table 2.6: Capital costs for selected reactor models

### 2.3.6. Commissioning Time

DRAFT

See observations in the previous section

I would apply a +0.5 years to planned commissioning time on the APR and AP based on the planned commissioning time of the EPR that differs by 0.5 between UK and Chinese plannings.

| **Reactor Model** | **Planned Commissioning Time** | **Last unit Commissioning Time** | Data Source |
|---|---|---|---|
| APR1400 | 5.0 in Korea | 10 and 8 in Korea | [8] |
| EPR 1750 | 5.0 in Europe, 4.5 in China | 9 in Taishan, 17 in Flamanville and Olkiluoto | [8] |
| AP1000 | 5.0 in USA | 9 in Sanmen, 10 and 11 at Vogtle | [8] |
| BWRX-300 | $2 \sim 3$ | NA | [67] |
| Rolls Royce SMR | 4.0 | NA | [68] |
| NuScale | 3.0 | NA | [69] |

Table 2.7: Technological readiness and planned commissioning time

### 2.3.7. Operational Costs

Lascks presentation of the attribute

Steigerwald et al. (2023) [63] provides comprehensive cost analyses for future nuclear installations, including capital and operational expenditures for both SMRs and large reactors. The study reports operational costs in $USD_{2020}/MWh_e$ for EPR and APR1400 designs, as well as a "Long-Term US" category which this analysis uses as AP1000 costs. For SMRs, operational costs were originally presented in power-based units ($USD_{2020}/MW_e$), requiring conversion to energy-based measurements ($USD_{2020}/MWh_e$) using a Gaussian distribution of capacity factors ($\mu = 90\%$, $\sigma = 5\%$) according to:

$$\frac{USD_{2020}}{MWh_e} = \frac{USD_{2020}}{MW_e} \cdot \frac{1}{CF \cdot 8760\frac{h}{year}}$$

The same study reports fuel costs of $6.23 USD_{2020}/MWh_e$ for PWR-based technologies with the exceptions of NuScale at $7.15 USD_{2020}/MWh_e$ and the only BWR reactor at $29.55 USD_{2020}/MWh_e$. For comparative context, operational cost estimates for utility-scale onshore wind, hydroelectric, and solar PV (with single axis tracking and DC battery storage) were obtained from the U.S. Energy Information Administration [62]. All cost data have been standardized to 2020 USD to maintain consistency with the primary data sources [63, 61], values are presented in Table 2.8.

| **Reactor Model** | **Operative Cost [$/MWh]** | Data Source |
|---|---|---|
| APR-1400 | 18.44 | [63] |
| EPR-1750 | 14.26 | [63] |
| AP-1000 | 18.69 | [63] |
| BWRX-300 | $18.30 \pm 0.82$ | [63] |
| Rolls Royce SMR | $65.68 \pm 2.94$ | [63] |
| NuScale | $22.76 \pm 1.02$ | [63] |
| **Other Power Plants** | | |
| Onshore Wind | 28.08 | [62] |
| Solar PV + Battery | 33.33 | [62] |
| Hydroelectric | 28.49 | [62] |

Table 2.8: Economic costs for selected reactor models

### 2.3.8. Potential Impact on the energy market

To write

### 2.3.9. Potential Impact on greenhouse gas emissions

To write

### 2.3.10. Waste

To write

### 2.3.11. Size - Nominal output power

To write

### 2.3.12. Load Following Capabilities

Load-following capability can be quantitatively assessed using characteristic curves that represent ramp-up and ramp-down rates as a function of load (expressed as percentage of nominal power $\%P_{nom}$), as illustrated in Figure 2.1. To facilitate comparative analysis, these performance curves are reduced to scalar values through integration of the area under said curve.

For benchmarking purposes, load-following capabilities were calculated for two modern gas turbine configurations: A heavy-duty combined cycle plant featuring two GE Vernova 9HA.02 gas turbines (2015 data) [70] and an equivalent plant configuration utilizing aeroderivative GE LM2500XPRESS+G5 DLE UPT gas turbines (2025 data) [71]. Hydroelectric plants are also included as a reference for the highest flexibility, a 1989 publication by the US Office of Technology Assessment [72] reports that the response rate of large (>60MW) hydroelectric plants ranges from 4 to 6 %$P_{nom}$/s, which translates to 240 to 360 %$P_{nom}$/min. A graph from a 1994 conference reported in a technical analysis from 2009 shows that some hydro plant can achieve 0 to 100% in much less then a minute and from black startup to full power in around 5 minutes For the comparison we indicate greater then 100%. [73, 74]

To establish an upper benchmark hydroelectric plants were included in the comparative analysis. A 1989 U.S. Office of Technology Assessment report [72] documents that large (>60 MW) hydroelectric facilities demonstrate remarkable load-following capabilities, with response rates ranging from 4 to 6 %$P_{nom}$/s (equivalent to 240-360 %$P_{nom}$/min). Additional performance data from a 1994 conference presentation, cited in a 2009 technical analysis [73, 74], indicates that certain hydroelectric installations can achieve 0-100% load changes in under one minute and black start to full power operation in approximately five minutes. For the purposes of this comparison, hydroelectric flexibility is conservatively represented as >100 %$P_{nom}$/min.

The results of these analyses are presented in Table 2.9.

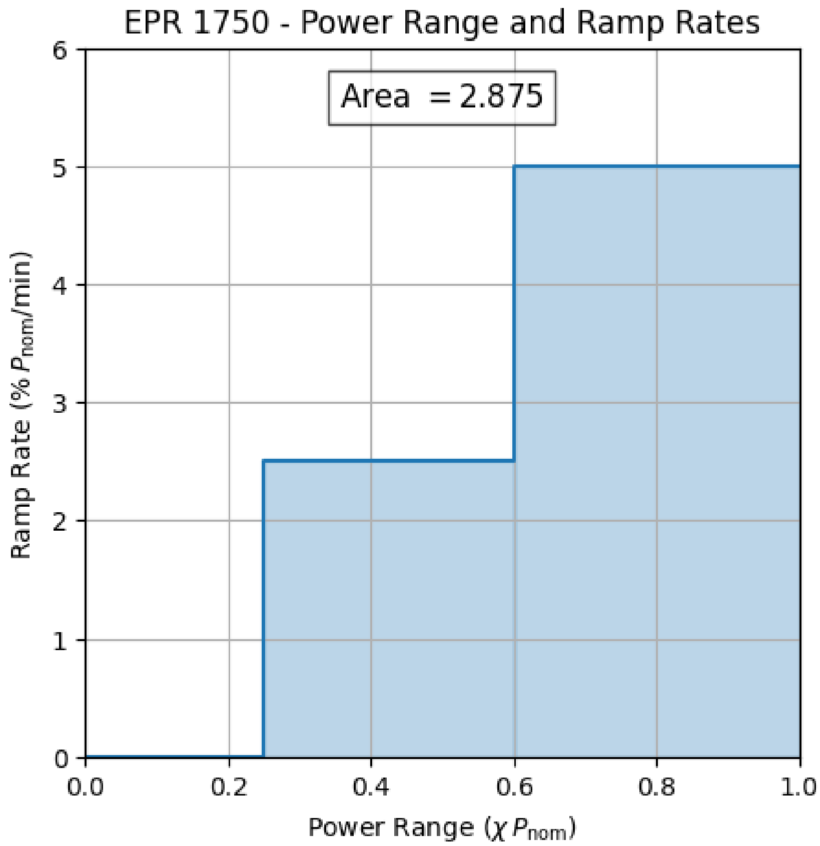


Figure 2.1: Example load following capability curve for EPR-1750

| **Reactor Model** | **Load Following Capability** $\left[\frac{\%P_{nom}^2}{min}\right]$ | Data Source |
|---|---|---|
| APR1400 | 0.21 | [45] |
| EPR 1750 | 2.88 | [75] |
| AP1000 | 4.25 | [45] |
| BWRX-300 | 0.25 | [45] |
| Rolls Royce SMR | 2.00 | [45] |
| NuScale | 2.00 | [76] |
| GE 9HA.02 2xCC | 7.02 | [70] |
| GE LM2500XPRESS+G5 2xCC | 40.54 | [71] |
| Hydroelectric | >100 | [72, 73, 74] |

Table 2.9: Load following capabilities and power output

# 3 Analysis

# 4 | Results

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# A | Appendix A

If you need to include an appendix to support the research in your thesis, you can place it at the end of the manuscript. An appendix contains supplementary material (figures, tables, data, codes, mathematical proofs, surveys, . . . ) which supplement the main results contained in the previous chapters.

# B | Appendix B

It may be necessary to include another appendix to better organize the presentation of supplementary material.

# List of Figures

# List of Tables

# List of Symbols

| **Variable** | **Description** | **SI unit** |
|---|---|---|
| $\boldsymbol{u}$ | solid displacement | m |
| $\boldsymbol{u}_f$ | fluid displacement | m |

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